
T-OCS V7.0 REVISION-2

NON-CONFIDENTIAL CORPORATE TECHNICAL & TECHNICAL-QUALIFICATION PRESENTATION

A controlled engineering conversation — not a sales claim, not a validation report, and not a manufacturing release.

L1 • NON-CONFIDENTIAL

Prepared for technical qualification meetings with ARRK Engineering, Calspan Automotive Test Services, and Toyota Boshoku America.

T-OCS

ENGINEERING QUALIFICATION

Process • Evidence • Milestones

How to use this meeting system

1 ALIGN

Define the engineering problem, the reason it matters, the functional concept, and the current evidence boundary.

The objective is shared technical understanding — not persuasion.

2 QUALIFY

Use C1–C5 to extract actual capability, required evidence, entry criteria, commercial structure, and NDA trigger from the other party.

Every answer is recorded as a factual meeting result.

3 CONVERT

Translate the meeting into a Technical Scope → Evidence → Entry Criteria → Action → Deliverable → Acceptance Criterion → Owner → Deadline chain.

HARD STOP

No “interested”, “potential partner”, or “good meeting” is treated as an engineering outcome without a measurable conversion.

The engineering problem

Dynamic occupant motion is a coupled system problem.

Occupant motion

Relative motion of the occupant must be considered together with seat and restraint behavior.

Kinematics

Seat translation and seatback motion determine how geometry, restraints and contacts evolve during the event.

Load path

Forces and reactions must have identifiable paths through the seat structure into the vehicle interface.

Energy / load management

The engineering question is how motion, force-time history and available stroke interact — not how to maximize force.

Structural response

Members, joints, interfaces and boundary conditions govern whether the intended response can physically occur.

Why rear-seat motion deserves disciplined evaluation: the occupant, restraint, seat structure and vehicle interfaces form a coupled dynamic system, so a promising local feature is not enough to establish system behavior.

Source basis: V6/V7 disclosure framing of occupant motion as coupled to seat + restraint + occupant, and explicit separation of architectural hypothesis from validated crash performance.

Why the problem matters

- Occupant motion is time-dependent and configuration-dependent.
- Seat motion can change restraint geometry and contact relationships during an event.
- Structural response can alter the actual motion that the occupant experiences.
- Any claimed improvement therefore needs a chain of evidence from geometry to dynamics to test correlation.

What the program must answer

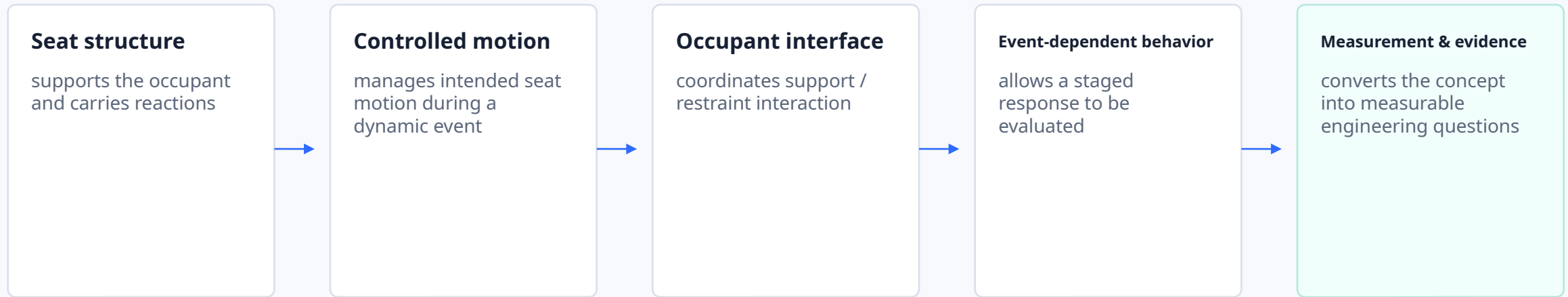
1. Can the intended motion occur without unintended interference?
2. Can the load paths be defined and instrumented?
3. Can component behavior be characterized independently?
4. Can the seat + restraint + occupant response be measured?
5. Can model predictions be correlated before any broader claim is made?

NO CLAIM OF PROVEN BENEFIT

The engineering case is: this is testable, falsifiable, and measurable — not yet proven.

Source basis: V7 evidence discipline and validation gates; current baseline explicitly excludes crash-safety validation claims.

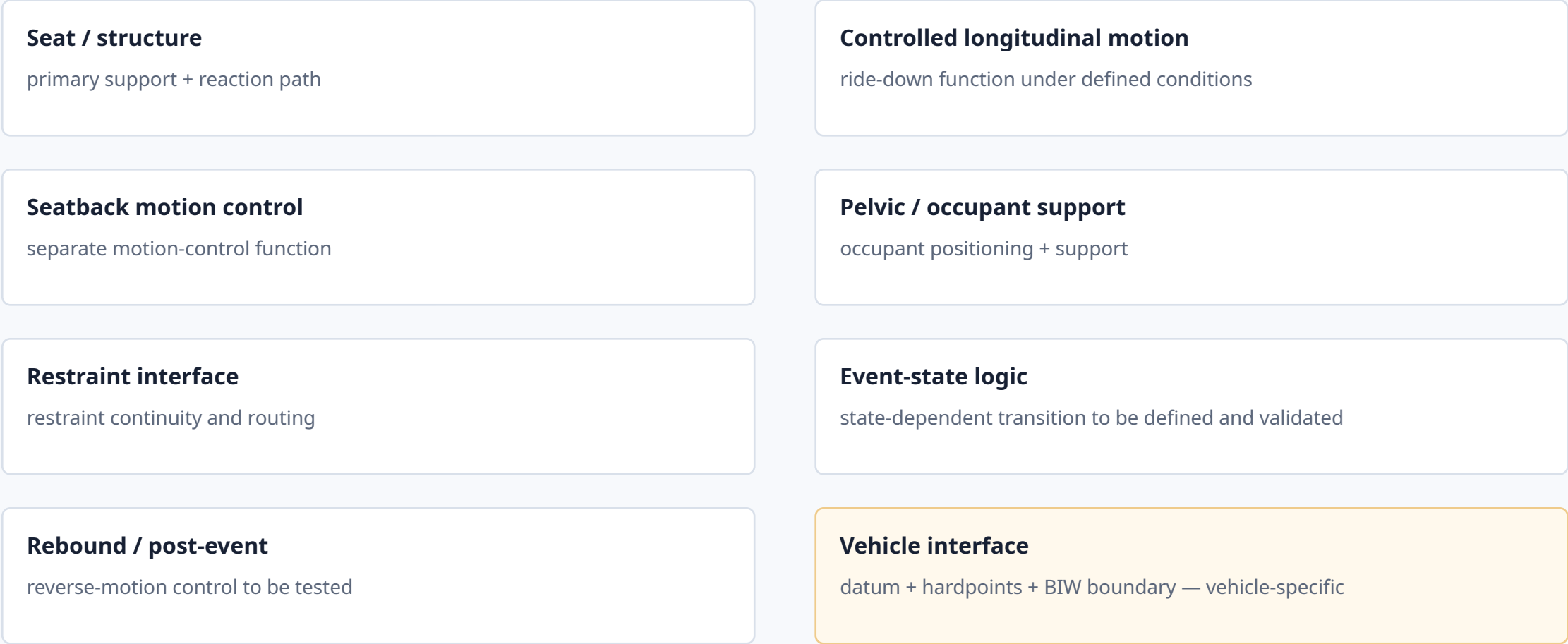
T-OCS V7 functional concept



L1 deliberately stops before proprietary mechanism disclosure.

No exact geometry • no CAD/STEP • no detailed force/energy assumptions • no patent-sensitive implementation details

System architecture — functional separation



Functional architecture only. Detailed implementation and exact part relationships are restricted beyond L1.

Why this approach is worth evaluating

1

Problem

Dynamic occupant motion couples kinematics, load paths, energy management and structure.

2

Design hypothesis

Coordinated seat motion and occupant-support functions may create a measurable alternative response pathway.

3

Functional approach

Separate the functions so each can be analyzed, measured and falsified.

4

Engineering rationale

A staged program reduces ambiguity: geometry → model → component behavior → dynamic test → correlation.

5

Required evidence

Only repeatable measured or independently verified evidence can support performance conclusions.

Intended controlled response

Motion

seat displacement / velocity / acceleration

Seatback

angle, angular rate, angular acceleration

Load path

left/right reactions and structural load transfer

Occupant

trajectory, pelvis/torso/head motion

Restraint

timing, routing, load history, continuity

Component behavior

force–displacement, hysteresis, temperature, cycles

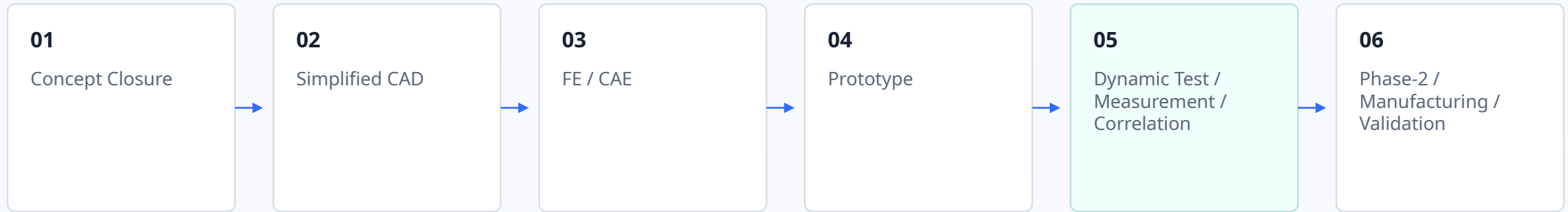
Rebound

reverse velocity, damping / control, secondary excursion

These are observables / evidence targets, not current T-OCS results.

V6/V7 disclosure explicitly maps claims to later observables; current V7 evidence package does not establish physical measurements.

Development path



Gate principle: existence of a previous stage is not evidence of completion of the next stage.

Each transition requires a defined artifact, owner, acceptance criterion, and evidence record.

Current engineering stage

CONTROLLED

- Functional architecture
- Engineering intent
- Requirements / evidence framework
- Engineering completion work order
- Development path
- Evidence-control discipline
- Current reconstruction / model-level engineering information

NOT YET PROVEN

- Physical validation / measurement
- Final manufacturability
- Final material closure
- Final fastener / joint closure
- Absorber characterization
- Lock / rebound validation
- Production readiness
- Crash / occupant-safety validation

Current V7 baseline: research / engineering reconstruction; manufacturing release and safety validation are not established.

Existing evidence vs open evidence

Architecture / problem definition	AVAILABLE	V7/V6 disclosure + controlled architecture
Engineering definition framework	AVAILABLE / PARTIAL	requirements, interfaces, normalization controls
Model-level kinematics	AVAILABLE AS MODEL EVIDENCE	not physical capability
Physical component behavior	OPEN	requires controlled tests
Vehicle datum / hardpoints	OPEN	vehicle-specific input required
Materials / fasteners / joints	OPEN	controlled supplier / engineering evidence required
Crash / sled / occupant results	NOT ESTABLISHED	no accepted T-OCS validation result
Manufacturing release	NOT ESTABLISHED	drawings / PMI / tolerances / process / inspection

The meeting asks for evaluation of the program and prerequisites — not acceptance of an unverified safety claim.

Source basis: current V7 executive report + master completeness audit.

Intended validation path



V6/V7 disclosures define this as an evidence ladder. The current package does not claim those later gates are completed.

Partnership models



No organization is represented here as a partner, client, sponsor, investor, or approved test provider unless the organization itself has documented that status.

Proposed Phase-1 structure

01	Scope	Define and challenge the technical feasibility questions within the agreed company scope.
02	Inputs	L1 package + only the additional inputs explicitly required by the selected provider.
03	Work	Engineering analysis / test planning / feasibility review as agreed.
04	Outputs	Named artifacts, documented findings, unresolved dependencies, and engineering decision recommendation.
05	Acceptance	Objective deliverable-based acceptance criterion agreed before start.
06	Boundary	No promise to close the entire V7 gap universe; no automatic manufacturing or safety conclusion.

Technical Qualification Framework C1–C5

C1 Technical Scope What can you actually do?	C2 Required Evidence What evidence do you require?	C3 Entry Criteria What must exist before Phase-1 starts?	C4 Commercial Structure How can the work be contracted and accepted?	C5 NDA Trigger What is L1, what needs NDA, what stays restricted?
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Close every meeting by converting answers into: Technical Owner → Next Action → Deadline.

Expected questions — controlled answer pattern

Every material answer uses:

KNOWN

LIMITATION

EVIDENCE NEEDED

NEXT ACTION

What exact problem does T-OCS solve?

It targets coordinated control of occupant motion, seat motion and load transfer; it is not presented as a proven safety outcome.

What is proven today?

Architecture, engineering framework and model-level evidence exist; physical performance and safety validation are not established.

Do you have native CAD?

The current controlled external representation is STEP-based assessment geometry; no native CATIA V5 master is asserted in the available baseline.

What would make you abandon the concept?

A controlled falsification result showing the intended response cannot meet pre-agreed technical criteria.

What is the cheapest credible falsification path?

A narrowly scoped, measurable engineering test that can discriminate the core hypothesis before broader investment.

Adversarial questions — answer without hand-waving

Why should we participate?

Because the program presents a bounded, testable technical question with explicit evidence gates — not because the outcome is assumed.

What is not proven?

Physical validation, final manufacturability, material/joint closure, component characterization, crash performance and production readiness.

What do you need from us?

Only the scope-specific expertise, evidence or test capability that the provider explicitly agrees to supply.

What is the IP position?

L1 excludes proprietary mechanism and patent-sensitive detail; controlled disclosure is gated by NDA and project authority.

How do we know this is not just paper engineering?

By requiring falsifiable acceptance criteria, independent measurements and documented correlation before performance claims.

Problem-solving workshop

Requirement Conflict

Conflict → competing interpretations → evidence needed → decision criteria

Packaging / Interface Conflict

Observed interface issue → geometry ownership → vehicle dependency → test / review

Kinematic Feasibility

Motion intent → joint assumptions → clearance sweep → falsification

Load / Energy Uncertainty

Unknown law → candidate models → component evidence → acceptance

Testability / Correlation

Unmeasurable quantity → instrumentation plan → model observables → correlation rule

WORKSHOP CLOSE

Problem → Hypotheses → Constraints → Alternatives → Evidence → Falsification → Decision Criteria → Owner → Deadline

Brainstorming + team-building control

A Compatible Improvement

B Architecture Modification

C Design-Intent Change

D IP / Disclosure Conflict

Every new idea must become a controlled record

Idea → Technical Impact → Evidence → Decision → Owner → Deadline

One task = One accountable owner = One deadline

Any proposed geometry change stays PROPOSED until separately reviewed and authorized.

No live brainstorming item becomes an authoritative V7 change by discussion alone.

Company-specific qualification paths

ARRK Engineering

Passive Safety → CAD / engineering closure → CAE / FE → prototype / test planning → acceptance criteria

Primary qualification output: technical scope, required inputs, deliverables, acceptance gate, and NDA trigger.

Calspan Automotive Test Services

Dynamic test → instrumentation → measurement → correlation → test preconditions → cost / schedule

Primary qualification output: candidate test scope, test article prerequisites, measurement outputs, quotation structure, and actual sponsorship / co-development availability if offered.

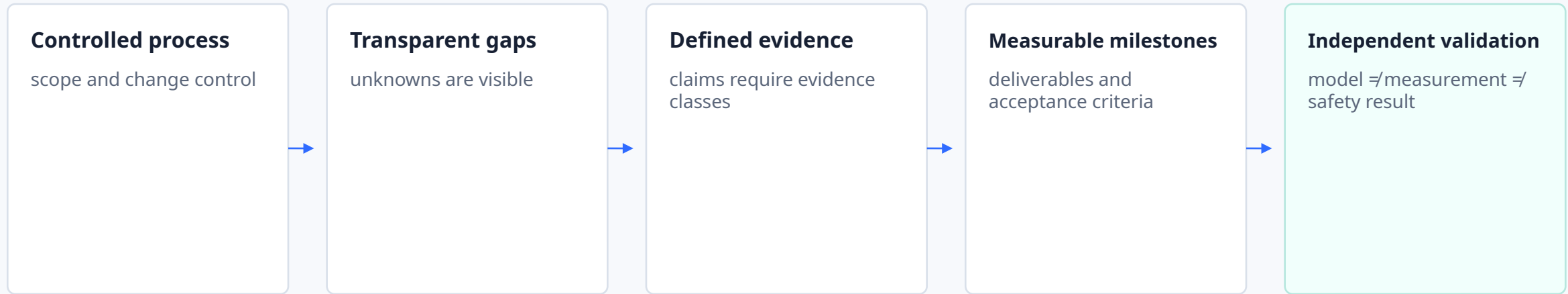
Toyota Boshoku America

Seat advanced development → technical fit → engineering contribution → co-development → milestones → NDA → internal technical owner

Primary qualification output: feasibility criteria, required inputs, milestone definition, NDA trigger, governance and next decision point.

Organization-specific historical communications remain unverified where the referenced contact source is not available in the current runtime. Do not infer prior commitments.

Why trust the program process?



The ask is not “trust that the technology works.”

The ask is “evaluate whether the engineering program is rigorous enough to prove or falsify the hypothesis.”

DECISION / NEXT STEP

Close the meeting with one agreed engineering decision — or an explicit OPEN item.

Technical Scope		Required Evidence	
Entry Criteria		NDA Status	
Next Action		Owner	
Deadline		Next Review	

STOP CONDITION: If a field cannot be supported by the meeting record, mark UNKNOWN / OPEN / PENDING HUMAN RESPONSE.